Revision 4 Memo

2.1

2.1.1 $f(x) = \frac{a}{x} + q$

$Asimptote: y = -1$

$f(x) = \frac{a}{x} - 1$

$Sub: (3; 0)$

$0 = \frac{a}{3} - 1$

$1 = \frac{a}{3}$

$1 \times 3 = a$

$3 = a$

$\therefore \quad f(x) = \frac{3}{x} - 1$

$g(x) = mx + c$

$y - intercept; (0; 1)$

$g(x) = mx + 1$

$Sub: (3; 0)$

$0 = m(3) + 1$

$-1 = 3m$

$-\frac{1}{3} = m$

$\therefore g(x) = -\frac{1}{3}x + 1$

2.1.2 $y \in R; y \neq -1$

2.1.3 $x \in R$

2.1.4 $h(x) = -f(x) + 3$

$= -\left(\frac{3}{x} - 1\right) + 3 \quad = -\frac{3}{x} + 1 + 3$

$= -\frac{3}{x} + 4$

2.2

2.2.1 $Turning\ point\ (0; -9)$

2.2.2 $y \in [-9; \infty)$

2.2.3 $f(x) = x^2 - 9$

$f(-2) = (-2)^2 - 9 = 4 - 9 = -5$

$g(x) = -x + 4$

$g(-2) = -(-2) + 4 = 2 + 4 = 6$

$\therefore BC = 6 - (-5) = 11\ units$

2.2.4 $-3 < x < 3; x \in R$

2.3

2.3.1 $y = 3$

2.3.2 $y \in \infty; y \neq 3$ or $y \in R - \{3\}$

2.3.3 $f(x) = 3^x + 3$

$sub\ x = 0\ (y - intercept)$

$f(x) = 3^0 + 3 = 1 + 3 = 4$

$\therefore A\ (0; 4)$

2.3.4 $x > 0$